

Pattaraphon Kenny Wongchamcharoen

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EDUCATION	University of California, Berkeley B.S. Industrial Engineering & Operations Research Minors in Data Science and Mathematics GPA: 3.97/4.00	Expected May 2027
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RESEARCH INTERESTS	Methods: Stochastic Modeling and Queueing Theory; Optimization; Game Theory; Statistical and Empirical Methods; Applied Machine Learning Applications: Human-AI Operations; AI Reliability in Operations; Service and Transportation Systems; Platform Operations and Marketplace Design
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PUBLICATIONS & WORKING PAPERS	[1] Do Large Language Models (LLMs) Understand Chronology? Wongchamcharoen, P. K., & Glasserman, P. <i>Proceedings of the 2026 AAAI Conference on Artificial Intelligence (AAAI-26)</i> Oral – Student Abstract & Poster Program (Top 11%) Poster – <i>AI4TS: AI for Time Series Analysis (AAAI-26 Workshop)</i> [proceedings] [arXiv] [code] [2] CentaurBench: Benchmarking LLM Capabilities on Augmenting vs. Automating Real-World Work Tasks Wongchamcharoen, P. K., Gulati, K., Fong, M. M., & Nagaraj, A. <i>NBER Working Paper #35663</i> [nber] [arXiv] [project dashboard] [3] On-Off Systems with Strategic Customers Sun, Y., Wongchamcharoen, P. K., Liu, Z., & Yan, C. <i>Revise and Resubmit (R&R) at Operations Research</i> [ssrn] [4] Flex or Fast? Incentive-Compatible Demand Allocation in Destination-Mode Ride-Hailing Networks <i>In preparation</i>
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OTHER RESEARCH PROJECTS	[1] Data-Driven Evaluation of Board of Directors Effectiveness Wongchamcharoen, P. K., Stringer, C., Hwang, B., Liu, I., & Li, K. <i>Best Data Visualization, Berkeley CDSS Data Discovery Symposium 2025</i> [poster] [2] Mixed-Integer Linear Program for Options Pricing and Portfolio Optimization Bin Abdulla, Q. M., Wongchamcharoen, P. K., Jamari, A., & Lee, J. <i>1st Runner-up, Wells Fargo & IEOR Bay Area Decision Sciences Summit 2025</i> [poster] [code] [3] Calyber: A Ridesharing Game Shen, Y., Yan, C., & Yan, J. <i>INFORMS Transactions on Education</i>, 26(3):218–225, 2026 Research assistant contribution; Runner-up, 2025 INFORMS Case Competition [article] [award]
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TALKS & PRESENTATIONS

Sun, Y., **Wongchamcharoen, P. K.**, Liu, Z., and Yan, C. (2026). “On-Off Systems with Strategic Customers.” Scheduled invited presentation at the *2026 INFORMS Annual Meeting, MSOM Invited Session* in San Francisco, CA.

Wongchamcharoen, P. K., Gulati, K., Fong, M. M., and Nagaraj, A. (2026). “CentaurBench: Benchmarking LLM Capabilities on Augmenting vs. Automating Real-World Work Tasks.” Presentation at the *2026 Wharton Generative AI & Business Conference* in San Francisco, CA.

Wongchamcharoen, P. K. (2026). “AI Is Getting Smarter... Are We Cooked?: Research Perspectives on the Capabilities, Limits, and Future of AI.” Invited talk at the *Berkeley Engineering Summer Machine-Learning & AI Research Training (BeSMART) Program* in Berkeley, CA. [slides]

Wongchamcharoen, P. K. and Glasserman, P. (2026). “Do LLMs Understand Chronology?” Presentation at the *2026 IISE Annual Conference* in Arlington, TX.

Wongchamcharoen, P. K. and Glasserman, P. (2026). “The Temporal Blind Spots of AI: Implications for IEOR Applications.” Presentation at the *Berkeley IEOR Annual Advisory Board Meeting* in Berkeley, CA. [press]

Wongchamcharoen, P. K. and Glasserman, P. (2026). “Do LLMs Understand Chronology?” Oral presentation at the *40th Annual AAAI Conference on Artificial Intelligence* in Singapore. [poster] [talk]

RESEARCH EXPERIENCE

UC Berkeley Industrial Engineering & Operations Research Berkeley, CA
Research Assistant

Advisor: Prof. Chiwei Yan

Jan 2024 – Present

- Extended the on-off dynamic queueing model with strategic customers to multi-server systems; reformulated the nonconvex throughput problem as a linear program and proved theoretical results on throughput scaling, demand saturation, and optimal service allocation under exogenous on-off policies, including a closed-form linear-scaling threshold via an explicit dual certificate; contributed to the revision of *On-Off Systems with Strategic Customers* (R&R, *Operations Research*).
- Designed and validated a discrete-event simulator with Poisson arrivals, strategic customer joining, exhaustive service, and counterfactual wait tracking; developed learning dynamics and algorithms for endogenous equilibrium computation; evaluated stochastic-to-fluid convergence and out-of-sample policy performance.
- Estimated rider patience and recovered suppressed demand using a hierarchical Bayesian mixed-effects model with Bay Area Rapid Transit (BART) Yellow Line origin–destination and GTFS data; optimized reusable dwell-time policies across 48 station-direction queues using sample-average approximation, projecting peak-period throughput gains of up to 24%; invited to present in an MSOM session at the 2026 INFORMS Annual Meeting and contributed to securing a \$40,729 UC ITS research grant for passenger-centric BART schedule optimization.
- Deployed *Calyber*, a 2025 INFORMS Case Competition runner-up, in a graduate supply-chain course; developed Python notebooks and teaching materials; guided teams in designing pricing and matching policies using a dataset of more than 10,000 Chicago rider arrivals, supporting the case’s subsequent publication in *INFORMS Transactions on Education*; built an AI-agent competitor to benchmark agentic policies against student teams on profit, conversion, and matching.

UC Berkeley Haas School of Business

Berkeley, CA

*Research Assistant, Data Innovation and AI Lab (DIAL)**Advisor: Prof. Abhishek Nagaraj*

Jan 2024 – Present

- Led a four-person research team to conceptualize and develop *CentaurBench*, a benchmark distinguishing an LLM’s ability to automate professional work from its ability to augment another agent.
- Designed and ran experiments across seven professional tasks and ten frontier models, finding that automation and augmentation rankings diverge; NBER Working Paper; presented at the 2026 Wharton Generative AI & Business Conference.
- Engineered a multi-provider asynchronous Python evaluation pipeline across six model providers; built a reproducible LLM-as-judge harness using anonymized round-robin comparisons, randomized response ordering, micro-rubrics, four-model judge panels, and *leave-family-out* masking; validated 15,000+ scored comparisons using rank correlations, inter-judge agreement, Krippendorff’s alpha, and sensitivity analysis; created a public reproducibility dashboard and repository auditing outputs, rankings, rationales, and replication diagnostics.
- Supported research on generative AI and online platforms, building logistic regression and random forest classifiers to distinguish human from AI-generated OpenStreetMap contributions.

Columbia Business School, Decisions, Risk, and Operations

New York, NY

*Summer Research Fellow**Advisor: Prof. Paul Glasserman*

Jun 2025 – Aug 2025

- Selected as 1 of 15 interns from more than 1,000 applicants for Columbia Business School’s Summer Research Internship Program to investigate whether large language models (LLMs) understand chronology.
- Built and deployed a reproducible Python pipeline using GPT-4.1, o3, Claude 3.7 Sonnet, and other frontier LLM APIs to simulate event-ordering tasks and evaluate chronological reasoning at scale across increasingly complex task families: basic sorting, conditional sorting, and anachronism detection; developed a statistical evaluation framework comparing reasoning models using Spearman rank correlation, Kendall rank correlation, and complementary rank-based diagnostics across model families and task configurations.
- Published the findings in the AAAI-26 proceedings and presented them orally in the Student Abstract Program (top 11%) and as a poster at the AI4TS: AI for Time Series Analysis Workshop; invited to present at the 2026 IISE Annual Conference and Berkeley IEOR Annual Advisory Board Meeting; cited in OpenAI’s 2026 paper, “Scaling social science research” [*working paper*].

UC Berkeley School of Law

Berkeley, CA

*Student Researcher**Advisor: Prof. Frank Partnoy*

Jan 2025 – May 2025

- Scraped SEC DEF 14A board skills matrices; applied t-SNE clustering and predictive modeling to construct data-driven measures of board effectiveness.

UC Berkeley Haas School of Business

Berkeley, CA

*Research Assistant**Advisor: Prof. David Evan Harris*

Aug 2023 – Jan 2024

- Conducted literature review on AI ethics and governance using Zotero.

PROFESSIONAL EXPERIENCE	Beats by Dre , Data Analytics Extern, Culver City, CA Jun–Aug 2024 Xcube Capital , Product Development & Analytics Intern, Singapore May–Aug 2024 MENTOR Global Consultants , Technical Product Mgmt. Intern Jan–Apr 2024 GUESS, Inc. , Strategy Contract Consultant Aug–Dec 2023
HONORS AND AWARDS	Tau Beta Pi Scholarship (Tau Beta Pi Association; \$2,000) 2026 Bonder Foundation Scholarship (INFORMS; \$400) 2026 Dean’s Lists (UC Berkeley College of Engineering) 2023–2026 CBS Research Internship Grant (Columbia Business School; \$7,500) 2025 Best Poster: Data Visualization Award (UC Berkeley CDSS) 2025 1st Runner-up, Bay Area Decision Sciences Summit (Wells Fargo; \$600) 2025 Cal Alumni Leadership Award Finalist (Cal Alumni Association) 2024 College of Engineering Mentorship Grant (UC Berkeley; \$850) 2024 ASEAN Scholarship (Singapore Ministry of Education; full 4-year award) 2018
TEACHING EXPERIENCE	Tutor & Grader , University of California, Berkeley Fall 2024–Present • IEOR 145: Fundamentals of Revenue Management (Prof. Rajan Udwani) • IEOR 142A: Machine Learning & Data Analytics I (Prof. Phillip Kerger) • IEOR 172: Probability & Risk Analysis for Engineers; IEOR 120: Principles of Engineering Economics (Prof. Daniel Pirutinsky) • CS 88: Computational Structures in Data Science (Prof. John DeNero) • Tutor for lower- and upper-division courses in mathematics, data science, computer science, and IEOR, Berkeley’s Center for Access to Engineering Excellence (CAEE) Primary Lecturer , Berkeley Engineering GLOBE Center Fall 2025–Present • Developed and taught <i>Foundations of Machine Learning</i>, an introductory machine-learning course for underserved Bay Area high-school students (Program Advisor: Prof. Phillip Kerger). [course website] Teaching Assistant , University of California, Berkeley Spring–Summer 2026 • IEOR C253: Supply Chain & Logistics Management (Prof. Chiwei Yan) • Berkeley Summer Machine Learning & AI Research Training Program (BeSMART); mentored student machine learning research projects and presentations, created assignments, and guest lectured on coding and agentic AI (Prof. Phillip Kerger)
SERVICE AND LEADERSHIP	Institute of Industrial & Systems Engineers at Berkeley Senior Advisor Fall 2025 – Present Director of Programs and Events Fall 2023 – Spring 2024 Singapore Students Association at Berkeley Vice President of External Relations Spring 2024 – Fall 2024
PROFESSIONAL MEMBERSHIPS	Institute for Operations Research and the Management Sciences (INFORMS); Institute of Industrial & Systems Engineers (IISE); Tau Beta Pi Engineering Honor Society
SKILLS & LANGUAGES	Python (NumPy, Pandas, Matplotlib, scikit-learn, Pyomo, TensorFlow); Java; SQL; Apache Spark; LLM APIs (OpenAI, Anthropic, Google Gemini); HuggingFace; L ^A T _E X English (Fluent); Thai (Native); Mandarin Chinese (Elementary)